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Vitor Greati

I am a PhD student in the final months of my study. I work in theoretical computer science at the boundary of formal logic, universal algebra and computational complexity, drawing on tools from proof theory, algebraic logic, well-quasi-order theory and infinite state systems.

Education

2022–present	**PhD, Computer Science**, *University of Groningen*, Groningen, The Netherlands
Advisor	Prof. Dr. Revantha Ramanayake
2020–2022	**MSc, Computer Science**, *Federal University of Rio Grande do Norte*, Natal, Brazil
Advisors	Prof. Dr. João Marcos and Dr. Sérgio Marcelino
Final grade	5.0/5.0
2018–2019	**BSc, Computer Science**, *Federal University of Rio Grande do Norte*, Natal, Brazil
Advisor	Prof. Dr. João Marcos
Final grade	9.7/10 summa cum laude
2015–2018	**BSc, Information Technology**, *Federal University of Rio Grande do Norte*, Natal, Brazil
Advisor	Dr. Selan Santos
Final grade	9.6/10 summa cum laude
2011–2016	**Technician in Informatics**, *Federal Institute of Rio Grande do Norte*, Natal, Brazil
Advisor	Dr. Bruno Costa
Final grade	9.7/10 summa cum laude
2011–2015	**Technician in Web Programming**, *Federal University of Rio Grande do Norte*, Natal, Brazil

Publications (Logic and Theoretical Computer Science)

Note In computer science, conference publications are peer-reviewed and count as much as journal papers. Moreover, authors are usually listed in alphabetical order.

[1] A.R. Balasubramanian, Vitor Greati, and Revantha Ramanayake. *Hypersequent calculi have Ackermannian upper bounds*. LICS 2026 (to appear). 2026. arXiv: `2602.19229 [cs.LO]`.

[2] Vitor Greati and Revantha Ramanayake. "Tight length theorems for multiset extensions of Higman's lemma". In: *Theoretical Computer Science* 1057 (2025), p. 115546. ISSN: 0304-3975. DOI: `10.1016/j.tcs.2025.115546`.

[3] Vitor Greati et al. "Analytic calculi for logics of indicative conditionals". In: *Automated Reasoning with Analytic Tableaux and Related Methods*. Ed. by Gian Luca Pozzato and Tarmo Uustalu. Cham: Springer Nature Switzerland, 2025, pp. 59–81. ISBN: 978-3-032-06085-3. DOI: `10.1007/978-3-032-06085-3_4`.

[4] Vitor Greati, Sérgio Marcelino, and Umberto Rivieccio. "Axiomatizing the logic of ordinary discourse". In: *Information Processing and Management of Uncertainty in Knowledge-Based Systems*. Ed. by Marie-Jeanne Lesot et al. Cham: Springer Nature Switzerland, 2025, pp. 390–405. ISBN: 978-3-031-74000-8. DOI: `10.1007/978-3-031-74000-8_32`.

[5] Vitor Greati et al. "Generating proof systems for three-valued propositional logics". In: *Handbook of Three-Valued Logics (to appear)*. 2025. arXiv: `2401.03274 [math.LO]`.

[6] Vitor Greati, Sérgio Marcelino, and Umberto Rivieccio. "Finite Hilbert systems for Weak Kleene logics". In: *Studia Logica* 112.6 (2024), pp. 1215–1241. ISSN: 1572-8730. DOI: `10.1007/s11225-023-10079-w`.

[7] Vitor Greati et al. "Adding an implication to logics of perfect paradefinite algebras". In: *Mathematical Structures in Computer Science* 34.10 (2024), 1138–1183. DOI: `10.1017/S0960129524000227`.

[8] Vitor Greati and Revantha Ramanayake. "Deducibility in the full Lambek calculus with weakening is HAck-complete". In: *Advances in Modal Logic, AiML 2024, Prague, Czech Republic, August 19-23, 2024*. Ed. by Agata Ciabattoni, David Gabelaia, and Igor Sedlár. College Publications, 2024, pp. 401–422. URL: `www.aiml.net/volumes/volume15/19-GreatiRamanayake.pdf`.

[9] Vitor Greati. "Hilbert-Style formalism for two-dimensional notions of consequence". MA thesis. Universidade Federal do Rio Grande do Norte, 2022. URL: `https://repositorio.ufrn.br/handle/123456789/46792`.

[10] Vitor Greati and João Marcos. "Finite two-dimensional proof systems for non-finitely axiomatizable logics". In: *Automated Reasoning*. Ed. by Jasmin Blanchette, Laura Kovács, and Dirk Pattinson. Cham: Springer International Publishing, 2022, pp. 640–658. ISBN: 978-3-031-10769-6. DOI: `10.1007/978-3-031-10769-6_37`.

[11] Vitor Greati, Sérgio Marcelino, and João Marcos. "Proof search on bilateralist judgments over non-deterministic semantics". In: *Automated Reasoning with Analytic Tableaux and Related Methods*. Ed. by Anupam Das and Sara Negri. Springer International Publishing, 2021, pp. 129–146. ISBN: 978-3-030-86059-2. DOI: `10.1007/978-3-030-86059-2_8`.

[12] Joel Gomes et al. "On logics of perfect paradefinite algebras". In: *Proceedings 16th Logical and Semantic Frameworks with Applications, LSFA 2021, Buenos Aires, Argentina (Online), 23rd - 24th July, 2021*. Ed. by Mauricio Ayala-Rincón and Eduardo Bonelli. Vol. 357. EPTCS. 2021, pp. 56–76. DOI: `10.4204/EPTCS.357.5`.

[13] Vitor Greati. "Hilbert calculi for the main fragments of Classical Logic". Undergraduate Thesis. Federal University of Rio Grande do Norte, 2019. URL: `https://repositorio.ufrn.br/items/f737862d-3ced-4c8d-bf7b-e7913c2246be`.

Preprints

[14] Amirhossein Akbar Tabatabai, Vitor Greati, and Revantha Ramanayake. *Predicative ordinal recursion on the constructive Veblen hierarchy*. Submitted to APAL. 2025. arXiv: `2510.18497 [math.LO]`.

[15] Nikolaos Galatos et al. *Complexities of well-quasi-ordered substructural logics*. 2025. arXiv: `2504.21674 [cs.LO]`.

Submitted

[16] Vitor Greati and Revantha Ramanayake. *Commutative full Lambek calculus with contraction and Kleene star is Π^0_1-complete*. Submitted to LICS 2026. 2026.

[17] Nikolaos Galatos et al. *Complexity bounds for commutative knotted substructural logics via proof theory*. Submitted to APAL. 2025.

[18] Nikolaos Galatos et al. *Lower complexity bounds for knotted substructural logics via counter machines*. Submitted to Transactions of the American Mathematical Society. 2025.

Publications (Data Science, Computer Vision, Material Sciences)

[19] Gilles Silvano et al. "Synthetic image generation for training deep learning-based automated license plate recognition systems on the Brazilian Mercosur standard". In: *Design Automation for Embedded Systems* 25.2 (2021), pp. 113–133. ISSN: 1572-8080. DOI: 10.1007/s10617-020-09241-7.

[20] Gilles Velleneuve Trindade Silvano et al. "Artificial Mercosur license plates dataset". In: *Data in Brief* 33 (2020), p. 106554. ISSN: 2352-3409. DOI: 10.1016/j.dib.2020.106554.

[21] André Fiel Borges et al. "Wearable optical device for the visually impaired in the classroom". In: *Research, Society and Development* 9.9 (2020). DOI: 10.33448/rsd-v9i9.7623.

[22] Roberto R. C. Lima et al. "Sodium-modified vermiculite for calcium ion removal from aqueous solution". In: *Industrial & Engineering Chemistry Research* 58.22 (2019), pp. 9380–9389. DOI: 10.1021/acs.iecr.9b00045.

[23] Aguinaldo Bezerra et al. "Enabling interactive visualizations in industrial big data". In: *IFAC* 53.2 (2020). 21st IFAC World Congress, pp. 11162–11167. ISSN: 2405-8963. DOI: 10.1016/j.ifacol.2020.12.292.

[24] Vinícius Ribeiro et al. "Brazilian Mercosur license plate detection: a deep learning approach relying on synthetic imagery". In: *2019 IX Brazilian Symposium on Computing Systems Engineering (SBESC)*. 2019, pp. 1–8. DOI: 10.1109/SBESC49506.2019.9046091.

[25] Aguinaldo Junior et al. "An industrial big data processing engine". In: *14º Simpósio Brasileiro de Automação Inteligente*. 2019. DOI: 10.17648/sbai-2019-111445.

[26] Vitor Greati et al. "Monitoring platform for vehicle irregularities in Smart Cities". In: *Congresso Brasileiro de Automática*. 2018 (in Portuguese).

[27] Vitor Greati et al. "Method for license plate recognition in uncontrolled environments". In: *Simpósio Brasileiro de Automação Industrial*. 2017 (in Portuguese).

[28] Vitor Greati, Vinícius Campos, and Ivanovitch Silva. "System based on automatic license plate recognition for vehicle monitoring and characterization of patrons in closed environments". In: *Escola Potiguar de Computação e suas Aplicações*. 2017 (in Portuguese).

[29] Vitor Greati et al. "A Brazilian license plate recognition method for applications in smart cities". In: *2017 IEEE First Summer School on Smart Cities (S3C)*. 2017, pp. 43–48. DOI: 10.1109/S3C.2017.8501395.

[30] Bruno Sielly Jales Costa et al. "A visual protocol for autonomous landing of unmanned aerial vehicles based on fuzzy matching and evolving clustering". In: *2015 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*. 2015, pp. 1–6. DOI: 10.1109/FUZZ-IEEE.2015.7337907.

Presentations

Invited talks

2025 Automated generation of Hilbert calculi for finite-valued logics. *Workshop Kleene logics and beyond*, Cagliari, Italy.

2022 Finite analytic two-dimensional proof systems for non-finitely axiomatizable logics. *Local seminar at the University of Bern*, Bern, Switzerland.

Conferences

2025 Analytic calculi for logics of indicative conditionals. *TABLEAUX'25*, Reykjavík, Iceland.

2025 Predicative recursion on the Veblen hierarchy. *Proof Society Workshop*, Ghent, Belgium.

2024 Deducibility in the full Lambek calculus with weakening is HAck-complete. *AiML'24*, Prague, Czech Republic.

2024 Axiomatizing the logic of ordinary discourse. *IPMU 2024*, Lisbon, Portugal.

2024 Knotted substructural logics have fast-growing complexity. *Logic Colloquium 2024*, Gothemburg, Sweden.

2023 Decidability and complexity upper bounds for weakly commutative knotted substructural logics. *MOSAIC 2023*, Vienna, Austria.

2023 Hilbert-style formalism for two-dimensional notions of consequence. *VCLA Awards*, Vienna, Austria.

2022 Finite analytic two-dimensional proof systems for non-finitely axiomatizable logics. *IJCAR 2022*, Haifa, Israel.

2022 Finite and analytic Hilbert-style systems for Paraconsistent Weak Kleene and Bochvar-Kleene logics. *Trends in Logic XXII*, online.

2021 Proof search on bilateralist judgments over non-deterministic semantics. *TABLEAUX 2021*, online.

2019 Towards the automated extraction of Hilbert calculi for fragments of classical logic. *19th Brazilian Logic Conference*, Joao Pessoa, Brazil.

2018 Platform for monitoring irregularities in vehicles in Smart Cities. *Congresso Brasileiro de Automática (CBA)*, Joao Pessoa, Brazil.

2017 A Brazilian license plate recognition method for applications in smart cities. *1st IEEE Summer School on Smart Cities*, Natal, Brazil.

2015 A visual protocol for autonomous landing of unmanned aerial vehicles based on fuzzy matching and evolving clustering. *FUZZ-IEEE 2015*, Istambul, Turkey.

Seminars

2026 Fast-growing complexities in substructural logics. *PhD seasonal seminar*, Groningen, The Netherlands.

2024 Deducibility in the full Lambek calculus with weakening is HAck-complete. *NetTCS*, Leiden, The Netherlands.

2024 Computational properties of non-classical logics. *AILo talk*, Groningen, The Netherlands.

Round tables

2025 Computational tools for logic and algebra, *Workshop Kleene logics and beyond*, Cagliari, Italy.

Teaching and dissemination

Teaching

2023-2026 **Computational complexity**, BSc course
- Description: Teaching assistant and lecturer (tutorials, assignments, grading, some lectures).
- Institution: Faculty of Science and Engineering, University of Groningen.
- When: Fall 2023, Fall 2024, Fall 2025.

2023-2026 **Algorithmic programming contests**, BSc course
- Description: Contest assistant.
- Institution: Faculty of Science and Engineering, University of Groningen.
- When: 2023, 2024, 2025.

2021–2021 **Mathematics for Computer Science III**, BSc course
- Description: Teaching assistant.
- Institution: Computer Science Department (DIMAP), UFRN.

2014–2015 **Programming contests in high school**, Institutional project
- Description: Tutor in the project "Training high school students for algorithm competitions", funded by CNPq-MEC.
- Institution: IFRN.

2012–2012 **Web Programming**, Technical course
- Description: Teaching assistant.
- Institution: Instituto Metropole Digital, UFRN

Dissemination

2019 **Fundamentals of Python and its applications in Chemistry**, Short course (4h), II Chemistry Week, UFRN

2019 **Breaking vulnerable CAPTCHAS with image processing techniques**, Short course (4h), Campus Party Natal 2

2016, 2017 **Introduction to LaTeX**, Short course (8h), Science and Technology Week, IFRN

Professional service

Supervision

2025 **Master's research internship**, *University of Groningen*
- Description: Co-supervisor of the MSc student Wouter Leferink in his intership program.
- Project: On a combinatorial proof for upper and lower bounds of bad sequences over $\mathbb{N}^k$.

2025 **Bachelor thesis**, *University of Groningen*
- Description: Co-supervisor of the BSc thesis of Scott Lantinga.
- Project: An interactive tool for exploring rule permutations in structural proof theory.

2024 **Bachelor thesis**, *University of Groningen*
- Description: Co-supervisor of the BSc thesis of Bogdan Bruma.
- Project: Boolean circuits in computational complexity: an accessible account (survey).

2023 **Bachelor thesis**, *University of Groningen*
- Description: Co-supervisor of the BSc thesis of Wouter Leferink.
- Project: Complexity classes beyond elementary (survey).

2023 **Bachelor thesis**, *University of Groningen*
- Description: Co-supervisor of the BSc thesis of Alexandra Stan.
- Project: Upper bounds for substructural logics using well-quasi-orders (survey).

Organization

2023 Head of the jury of the Groningen Algorithmic Programming Contest (GAPC).

2018 IEEE Symposium on Computers and Communications.

2016 IEEE Conference on Evolving and Adaptive Intelligent Systems.

2014 Science and Technology Week, IFRN.

Reviewer for journals

2025 International Journal of Approximate Reasoning (Elsevier)

2024 Logic Journal of the IGPL (Oxford Academy)

2024 Journal of Applied Non-Classical Logics (Taylor and Francis)

2024 Order (Springer)

2022 Journal of Logic and Computation (Oxford Academy)

2021 Studia Logica (Springer)

2021 Logic and Logical Philosophy (Nicolaus Copernicus University)

Member of assessment committees

2022 **Bachelor thesis defense**, *Department of Computer Engineering and Automation*, UFRN
- Description: Member of the assessment committee of the BSc defense of Mateus Abrantes.

2018 **Project assessment**, *V Exposition of Science and Technology of the North of Natal*, IFRN
- Description: Member of the assessment committee of research projects.

Experience in industry

2020–2022 **Researcher and developer**, *Logap Systems*, Natal, Brazil
- Description: Research leader and developer of the windfarm management system *Windbox*.

2015–2016 **Web Development Intern**, *IT Department*, UFRN
- Description: Web developer of the system SIGAA, used by the university as the main academic management system.

Grants and honours

Grants

2023 **E.W. Beth foundation**, *KNAW WF/2463-23*, Visits from external researchers and a three-day symposium on computational properties of substructural logics.

Honours

2023 **VCLA Outstanding Master Thesis Award – Runner-up**, *Vienna Center for Logic and Algorithms*, Vienna, Austria

2019 **2nd Best Paper Award**, *IX Brazilian Symposium on Computing Systems Engineering*, Natal, Brazil

2014 **Silver Medal**, *CRIA's Algorithm Cup*, Rio Info, Rio de Janeiro, Brazil

2014 **Bronze Medal**, *10th OBMEP*, IMPA, Brazil

2014 **Third best project of Natural Sciences and Mathematics**, *MOCITEC*, Federal Institute of Rio Grande do Norte, Natal, Brazil

2014 **Outstanding research award**, *MOCITEC IFRN*, ABRITEC, Natal, Brazil

2014 **Outstanding student award**, *Federal Institute of Rio Grande do Norte*, Natal, Brazil

Skills and competences

Languages

Portuguese — *Native*

English — *Fluent*

Programming languages

■■■■■ C++, Java, Python, LaTeX

■■■□□ Haskell, Scala

■■□□□ Lean, PHP

Software engineering, data science and others

■■■■■ Git — *CLI, GitHub, GitLab, CI/CD*

■■■■■ Linux — *Debian-based, Arch-based*

■■■■□	Docker	
■■■■■	Web frontend	*HTML5, CSS3, Javascript*
■■■■■	Web backend in Java	*Spring, Hibernate*
■■■■□	Web backend in Python	*django*
■■□□□	Web backend in PHP	*PHP Ignite*
■■■■■	Data science in Python	*pandas, scikit-learn, dask*
■■■■■	Databases	*MySQL, PostgreSQL, MongoDB*
■■■■□	Desktop programming	*Java*
■■■■□	Android programming	*Java*

Additional information

2020-2022 Student member of the Association for Symbolic Logic (ASL).

2019-2020 Volunteer in IEEE Computer Society at UFRN.